MZUZU UNIVERSITY

DEPARTMENT OF MATHEMATICS

EMT1201—CALCULUS 1

END OF SEMESTER EXAMINATIONS FOR ICT WEEKEND PROGRAM

January 5, 2016 **Time Allowed: 3 Hours**

8:00---11:00 am

Instructions

- *Answer all questions*
- *Marks for each question are indicated against each question*
- *Show your method clearly to receive full credit*
- *Calculators may be used where they are necessary*

1. Evaluate the following limits

   (a) $\lim_{x\to 2}\left(\dfrac{x^4-16}{x-2}\right)$ **(3 marks)**

   (b) $\lim_{x\to\infty}\dfrac{b+cx}{(d+5)x}$ where $b, c, d \in \mathbb{R}$ **(3 marks)**

   (c) $\lim_{x\to\infty}\dfrac{x^2+\sin x}{2x^2+\cos x}$ given that $|\sin x| \le 1$ **(4 marks)**

2. (a) What does it mean to say that a function $f(x)$ is continuous at a number $x = a$? **(2 marks)**

   (b) For what value of the number a is the function given by

   $$f(x) = \begin{cases} ax & if \quad x < 2 \\ 2x^2 & if \quad x \ge 2 \end{cases}$$ continuous? **(8 marks)**

3. Differentiate the function $f(x) = \sqrt{x}$ from the first principles. **(10 marks)**

4. A curve has the equation given by $x^2 + xy^3 + y^3 = 3$.

(a) Show that $\frac{dy}{dx} = -\frac{2x + 3x^2 y}{x^3 + 3y^2}$ **(5 marks)**

(b) Show whether the point (1, 1) lies on the curve or not? **(4 marks)**

(c) Find the equation of the tangent line to the curve at a point (1, 1). **(5 marks)**

5. Find all critical points of $f(x) = 3x^5 - 5x^3 + 2$ and classify all of them. **(20 marks)**

6. Find the maximum and minimum values of the function given by $f(x) = 2x^3 - 3x^2 - 12x + 1$ on the interval [-2, 3] and classify them as local maximum, local minimum, absolute maximum, or absolute minimum values **(20 marks)**

7. Compute the derivatives of the following functions

(a) $f(x) = x^2\{2x + 3)^3$ **(5 marks)**

(b) $f(x) = \frac{2x - 3}{x^3}$ **(6 marks)**

(c) $f(x) = \left(x - \frac{1}{x}\right)^2$ **(6 marks)**

End of Question Paper